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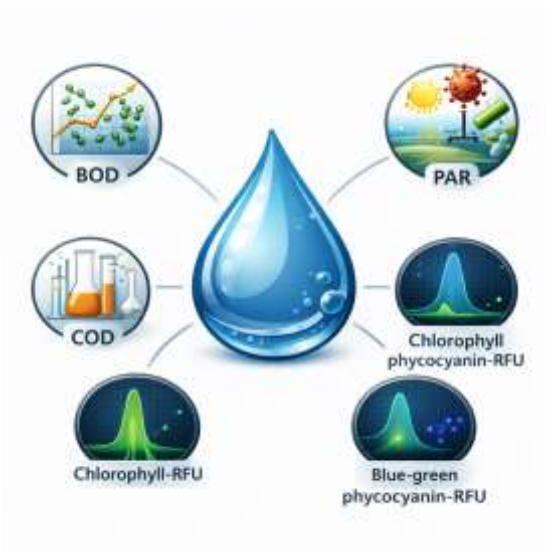
Soft sensor for water quality monitoring.

Recaps (Water Indicators)

Common measurement parameters



Specific measurement parameters

Recaps (Task)

Common measurement parameters

Specific one

Recaps (Dataset)

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Data in Brief

Volume 63, December 2025, 112290

Data Article

Dataset of high-frequency water quality and meteorological variables in Buffalo Pound Lake, Saskatchewan, Canada, 2014 – 2021

Helen Baulch [a 1], Jay Bauer [a c], Lisa Boyer [a], Katy Nugent [a], Kristin J. Painter [a], Jason J. Venkiteswaran [b], Lana Vuleta [a d], Colin J. Whitfield [a]

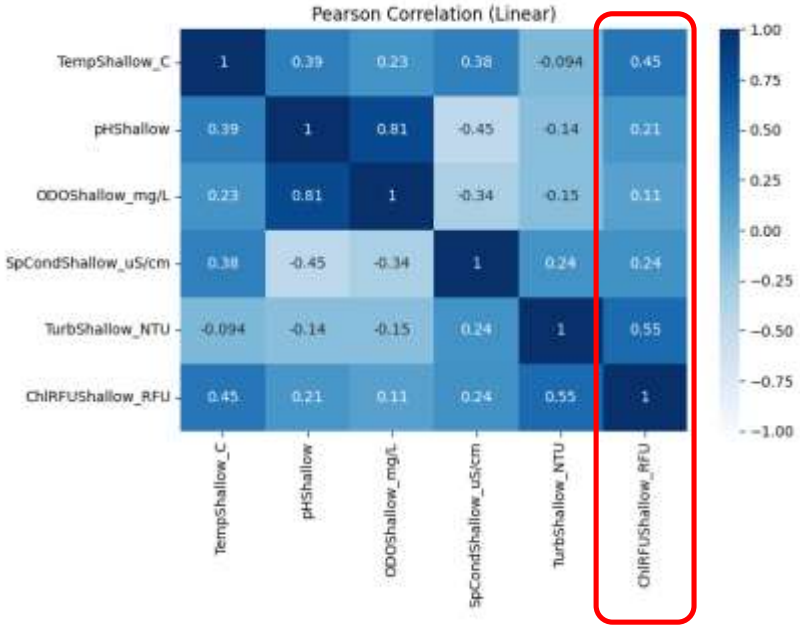
Recaps (Dataset)

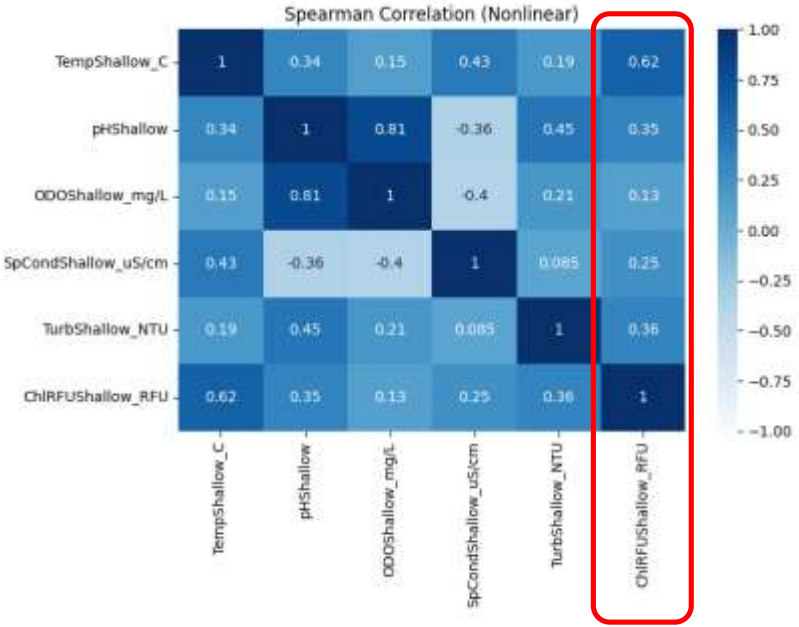
Common measurement parameters

Specific measurement parameters

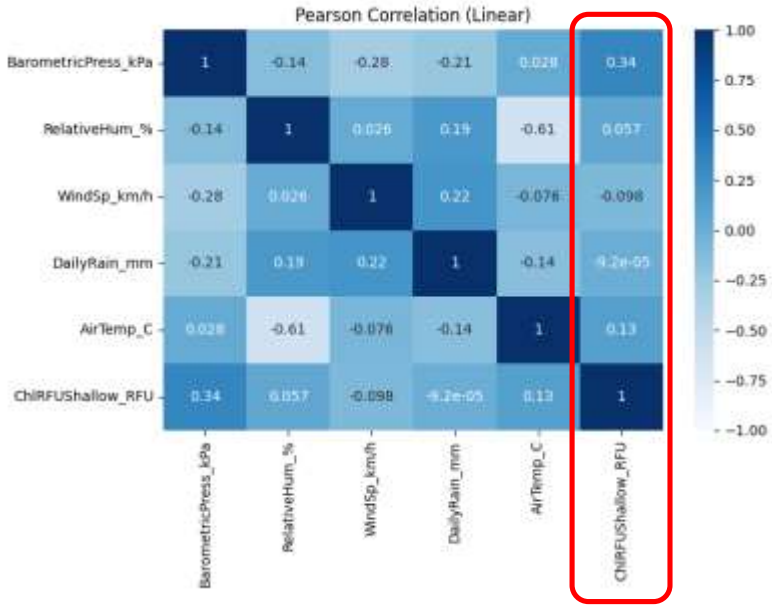
Meteorological parameters

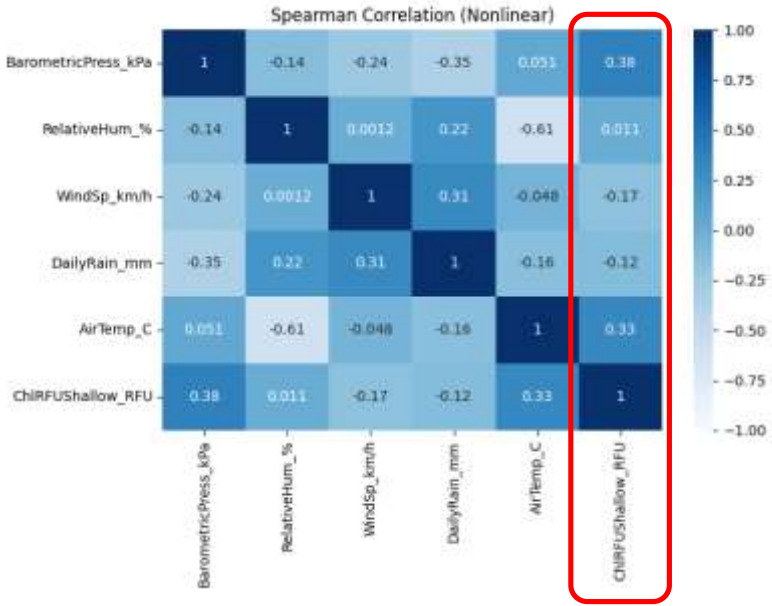
Recaps (Correlation – Water parameters)





Recaps (Correlation – Meteorological)





Recaps (Soft Sensor – Hard Calibration)

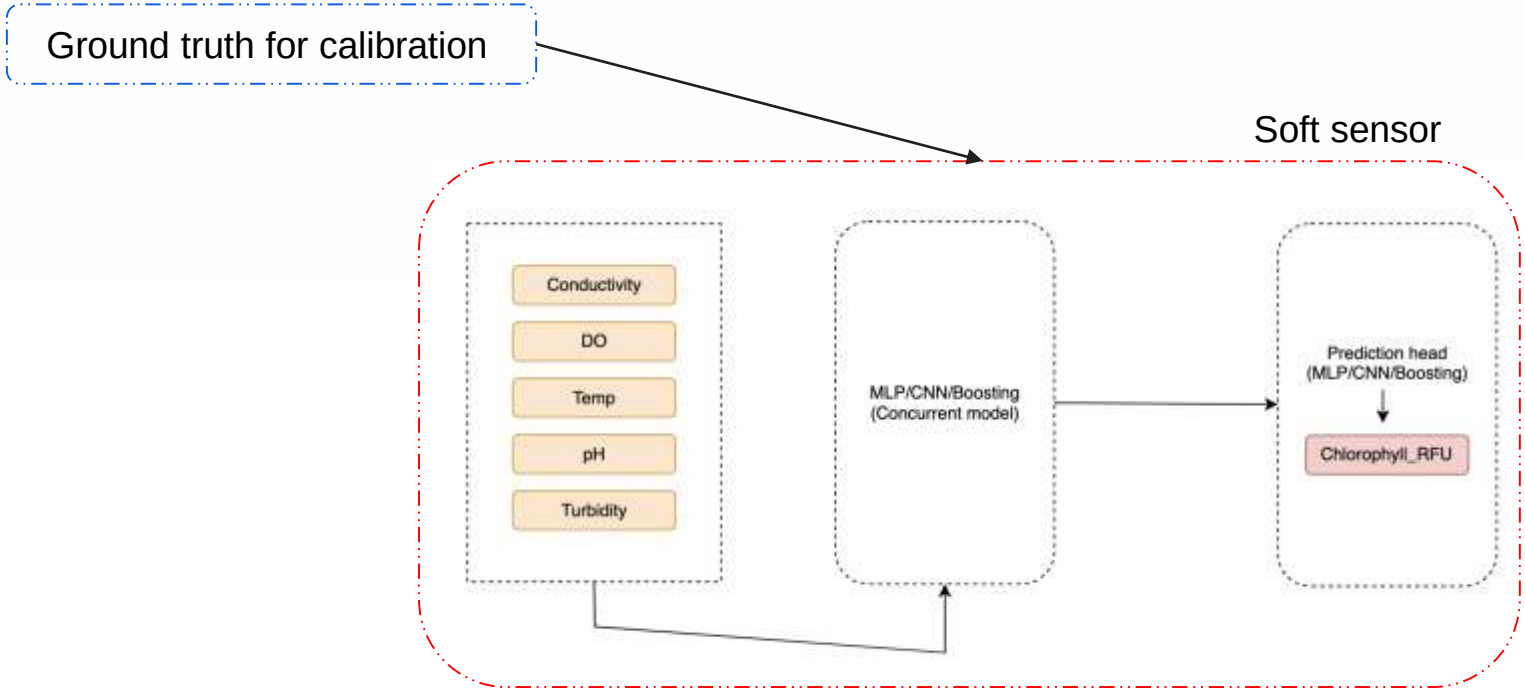
Specific ground truth

Specific prediction

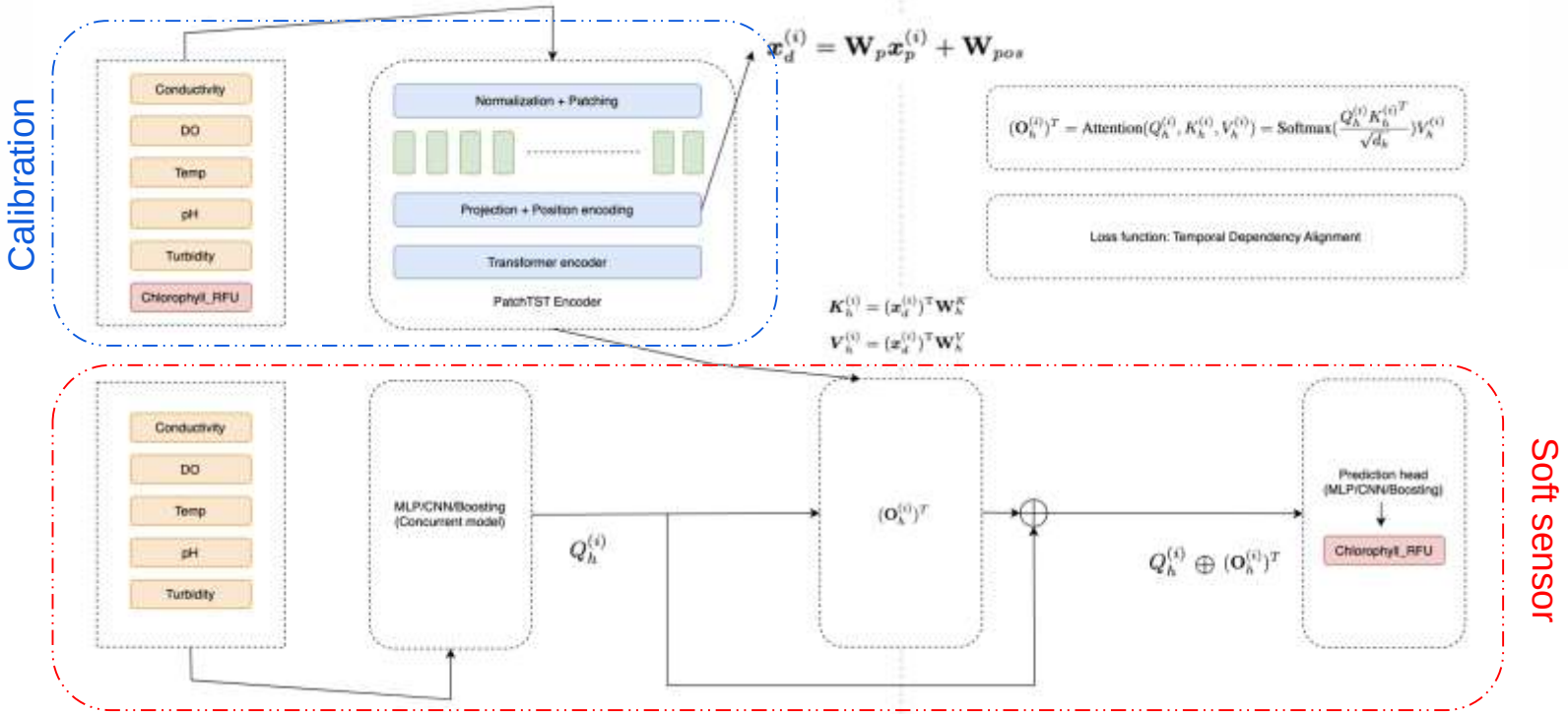
Inspire of periodic calibration of traditional sensor, soft sensor can be hard-calibration every single period.

Example: 14 days for prediction, 1 day for calibration.

Recaps (Soft Sensor – Hard Calibration)



Recaps (Soft Sensor – Hard Calibration)



Next steps (Soft sensor)

1. Create benchmark of this dataset
2. Find other datasets including BOD, COD, Bacteria …
3. Propose new calibration methods

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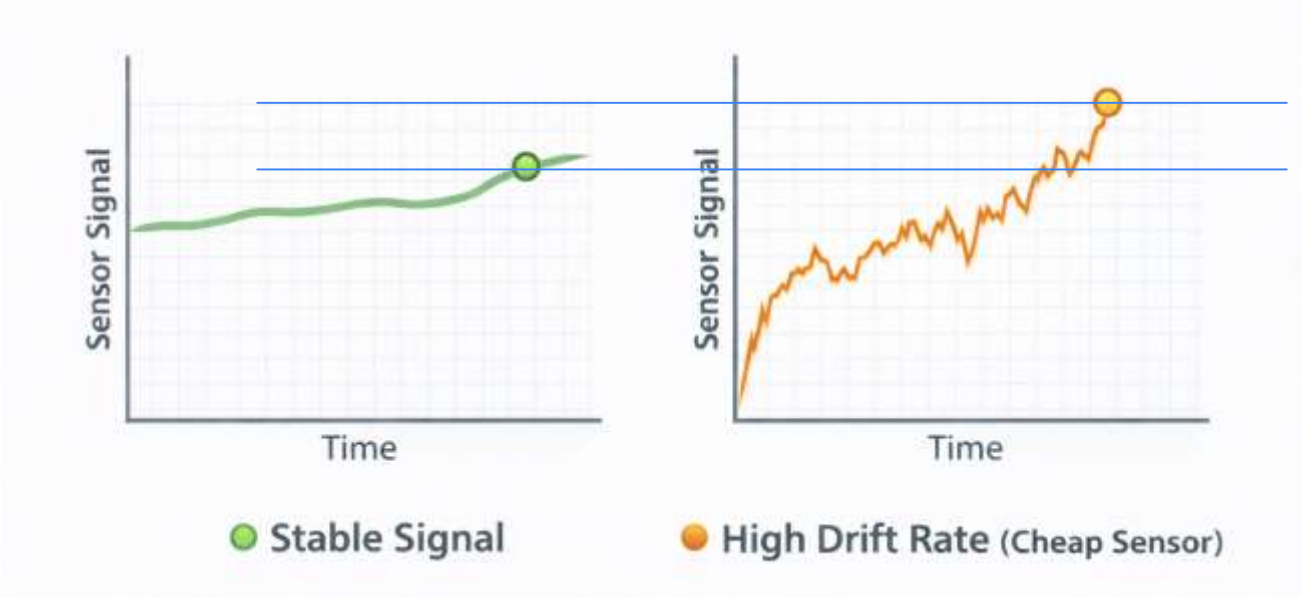
Soft calibration for water quality monitoring sensors.

Next steps (Soft calibration)

High drift rate sensors

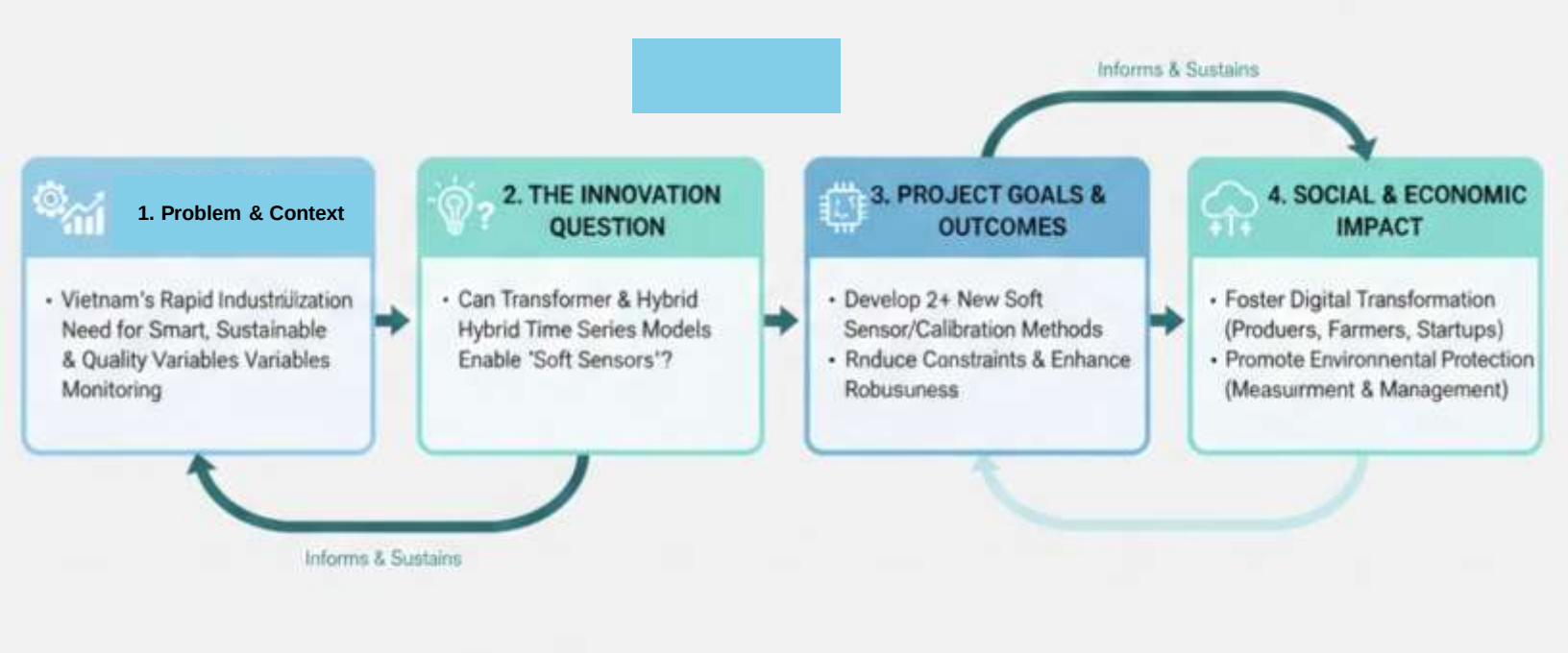
Low drift rate sensors

Next steps (Soft calibration)



compensation

Next steps (Soft calibration)



Problem & Context

- Vietnam's Rapid Industrialization.
- Need for Smart, Sustainable & Quality Variables Monitoring.

Innovation Question

- Can the development of ML/DL-based Long-term Time Series Forecasting models enable “Soft Sensors” ?

Goals & Outcomes

- Develop at least 2 new Soft Sensor, Soft Calibration methods.
- Reduce financial constrains & Enhance robustness.

Impacts

- Foster Digital Transformation (Producers, Farmers, Startups).
- Promote Environmental Protection